SVKM's NMIMS · Mukesh Patel School of Technology Management & Engineering, Mumbai

Immersive Virtual Reality & Augmented Reality (IVRAR - 702TG0C003) · AY 2026–27 Odd Semester

GROUP 13 TECHNICAL DOSSIER

Capstone ICA (20 marks): SBL Studio Project 10 · DBL Usability Case Study 5 · Studio Showcase & Viva Voce 5

Group / Batch	Group 13 · Batch B2 (as allotted in the finalised PBL titles sheet)
Monorepo folder	https://github.com/sunny-nanade/NMIMS-MPSTME-IVRAR-2026/tree/main/Group_13_Kartik_Soha_Atharv_Ojaswi
Submit (3 places)	(1) PDF on MS Teams, turned in by EVERY member · (2) this filled Word file AND its PDF committed to Group_13_Kartik_Soha_Atharv_Ojaswi/docs/IVRAR_Group_13_Technical_Dossier.docx / .pdf · (3) LinkedIn video URL in Section 5. Due Monday, 26-Oct-2026, 23:59 IST. Studio Showcase, Expo & Viva Voce: Saturday, 31-Oct-2026.
How to fill this file	Use Microsoft Word (desktop recommended). Choose answers from the 'Select ▾' drop-downs and type into the 'type here' fields. Do not delete or retype them, and do not edit this file in Google Docs: the drop-downs are how your answers are read.

Student name	Roll No.	SAP ID	Programme	Batch	Declared studio role
KARTIK AGRAWAL	I001	70122400060	B Tech (Artificial Intelligence)	B2	Biometric Authentication Lead
SOHA CHAND	I007	70122400036	B Tech (Artificial Intelligence)	B2	XR Systems Architect
ATHARV DIXIT	I013	70122400069	B Tech (Artificial Intelligence)	B2	Kinematic Telemetry Specialist
OJASWI GONDALIA	I019	70122400044	B Tech (Artificial Intelligence)	B2	Security QA & Threat Analyst

Batch column = official student list. Tell Dr. Nanade by 20-Oct if any name, ID or role is wrong.

SECTION 1 · Problem Bottleneck, Industry Standard & Aalborg Research Title *SBL Studio Project*

Authorised Aalborg title	How can continuous behavioral biometric authentication leveraging head and hand kinematic telemetry achieve equal error rates (EER) below 5% against avatar identity-spoofing in collaborative VR enterprise environments?
Original draft title	How can a simulated multi-factor authentication mechanism in Unity multiplayer VR prevent unauthorized avatar spoofing in virtual corporate boardrooms, where Verizon (2024) identifies XR-based meeting infiltration as an emerging attack vector and where research by Bailenson et al. (Stanford VHIL, 2023) demonstrates that avatar identity spoofing can extract confidential meeting content within 3.2 minutes of unauthorized session access?
Why the title changed	CRITICAL AI-HALLUCINATION PURGED: The submitted text cited 'Verizon 2024 XR Meeting Infiltration' and 'Bailenson Stanford VHIL 2023 3.2m data extraction'. Both citations were completely fabricated by an LLM! Naming fake DOIs/reports causes permanent desk rejection under COPE ethics rules.

1.1 Baseline friction & governing standard: claim verification table

Each statement below was carried over from your approved title sheet. Before you use it anywhere (dossier, video, paper), find the primary source yourself and record it, or correct or delete the claim. Standards: cite number, part and year. Statistics: cite the original study, not a blog or an AI answer.

Claim (as in title sheet)	Faculty pre-audit note	Primary source you verified (author, year, venue, DOI/URL)	Source type	Keep / Correct / Delete
ISO/IEC 19795 (Biometric Performance Testing).	*Not yet checked: verify*	ISO/IEC 19795-1:2021, Biometric performance testing and reporting, Part 1 (iso.org/standard/73515.html)	Standard (ISO / IEC / NFPA / NIST…)	Correct
Equal Error Rate: FAR(theta) = FRR(theta).	*Not yet checked: verify*	ISO/IEC 19795-1:2021 error-rate metrics; EER where FAR = FRR (our day2_eer.py)	Standard (ISO / IEC / NFPA / NIST…)	Keep
Legitimate research (Miller et al., IEEE TVCG 2020) proves unique head/hand kinematic movement signatures in VR.	*Faculty pre-audit, verified: Miller et al. (2020) is real but was published in Scientific Reports, not IEEE TVCG. Correct the venue: M. R. Miller, F. Herrera, H. Jun, J. A. Landay, J. N. Bailenson, 'Personal identifiability of user tracking data during observation of 360-degree VR video', Scientific Reports, 2020.*	Miller et al., Scientific Reports 10:17404 (2020), doi:10.1038/s41598-020-74486-y	Peer-reviewed paper (DOI)	Correct

1.2 Baseline friction in your own words (3–4 lines: who struggles, with what, how it is measured today):

In collaborative VR meetings, identity is checked once at the door. Afterwards, whoever puts on the headset inherits the avatar and sees confidential content, and nothing re-checks who is wearing it (0 identity checks per session after login).
The cheap 'continuous' safeguard often used instead, a head-speed anomaly detector, fails genuine users: on our own Quest build it would have revoked 11 of 12 genuine enrolments (pilot, 26-Oct) and 11 of 11 genuine sessions (team study, 27-Oct), typically within about 10 s.
Source: our on-device MFA and biometric logs (telemetry/real_participants, docs/paper). We measure success by EER, false-lockout rate and lockout latency.

SECTION 2 · C# Script Architecture & Mathematical / Behavioural Model *SBL Studio Project*

C# scripts in your monorepo folder today	KinematicTelemetryCollector.cs · ContinuousBiometricAuthManager.cs
Telemetry logger named in the title sheet	KinematicTelemetryLogger.cs
Core state machine	Enrolment → Template stored → Continuous verification window → Accept / Re-authenticate / Lock
Fundamental equations	FAR(θ) = impostor attempts accepted / impostor attempts FRR(θ) = genuine attempts rejected / genuine attempts EER = value where FAR(θ) = FRR(θ)
Toolchain (title sheet)	Unity 2022.3 LTS URP; C# KinematicTelemetryLogger.cs sampling 6-DoF head/controller position and velocity at 60Hz; Python distance classifier (scikit-learn Random Forest / Euclidean distance); EER and ROC curve generator.

2.1 Authorship of each script

Several scripts were provided as starter scaffolds. Be honest: mark what you wrote, what you changed, and what you used as given. Viva questions will target the code you claim.

Script	Written / Modified / Used as given	Main author (Roll)	What you changed or added (1 line)
KinematicTelemetryCollector.cs	Deleted / not used	Shared	Starter scaffold; replaced by KinematicFeatures.cs
ContinuousBiometricAuthManager.cs	Deleted / not used	Shared	Starter scaffold; replaced by ContinuousBiometricEngine.cs
ContinuousBiometricEngine.cs	Written by us	I001	
KinematicFeatures.cs	Written by us	I013	

2.2 Key code snippet (≤ 25 lines) that implements your state machine or main equation:

```
// Assets/Scripts/BoardroomXR/Biometrics/ContinuousBiometricEngine.cs, lines 208-236
208  private void ScoreStep(float[] f, float t) {
210      RawScore = Template.Score(f, LastZ);           // S = exp(-0.5 * mean(z^2))
214      if (!_smoothedInit) { SmoothedScore = RawScore; _smoothedInit = true; }
215      else SmoothedScore = Config.smoothing * RawScore + (1f - Config.smoothing) * SmoothedScore;
217      bool inGrace = t - _phaseStart < Config.graceSeconds;
218      if (SmoothedScore < Config.matchThreshold && !inGrace) {
220          Violations++;
221          if (Violations >= Config.violationsForLockout) {
223              SetState(BiometricState.LockedOut);      // host revokes IsVerified
225              OnLockout?.Invoke($"continuous biometric mismatch: score {SmoothedScore:F2} ...");
227              return;
228          }
229          SetState(BiometricState.Warning);
230      }
231      else {
233-235      Violations = 0; SetState(BiometricState.Verified); MaybeAdapt(f);  // adapt only when trusted
236      }
```

2.3 Equation derivation and one worked example with your own numbers and units:

Derivation.
Per window, z_i = clip((f_i - mu_i)/sigma_i, -6, 6) for the 16 features (mu, sigma from 60 s enrolment). Score S = exp(-0.5 * (1/16) * sum z_i^2), so S = 1 at the template mean and falls as the average deviation grows.
Smoothed score S~_t = 0.35*S_t + 0.65*S~_(t-1). A violation is S~ < theta. Balanced preset: theta = 0.15, k = 6 consecutive violations means lockout. EER: the theta at which FAR(theta) = FRR(theta).
Worked example (session S04, real headset data):
t = 78.38 s: raw S = 0.0020, so mean z^2 = -2 ln(0.0020) = 12.4, an average deviation of 3.5 sigma per feature (top: head speed, z = 6).
S~ = 0.35*0.0020 + 0.65*0.2212 = 0.1445 < 0.15, so violation 1 (Warning). Five more windows follow, and lockout comes at t = 80.90 s, 2.52 s after the first warning.
Where in the CSV: telemetry/real_participants/S04_window_scores.csv, columns t, raw_score, smoothed_score, violations, head_speed_mean.

30-second whiteboard rule: if a team member cannot explain an equation or code block on a whiteboard in 30 seconds without notes, simplify it or remove it. It will be asked in the viva.

SECTION 3 · GitHub Repository Verification & Git LFS Audit *SBL Studio Project*

Repository (monorepo)	https://github.com/sunny-nanade/NMIMS-MPSTME-IVRAR-2026
Your group folder	Group_13_Kartik_Soha_Atharv_Ojaswi/
Filled dossier location	Group_13_Kartik_Soha_Atharv_Ojaswi/docs/IVRAR_Group_13_Technical_Dossier.docx and .pdf
Release tag	group-13-v1.0 · build: Quest .apk or PCVR .exe + Python classifier

Final commit hash (full 40 characters)	6b5e67742f271079066baaf14eafe516f411e118
Release build link (.apk / .exe)	https://github.com/kartik-agrawal-01/NMIMS-MPSTME-IVRAR-2027/releases/tag/group-13-v1.0 (PR: github.com/sunny-nanade/NMIMS-MPSTME-IVRAR-2026/pull/28)
Root .gitattributes tracks our binary asset types via Git LFS	Yes
`git lfs ls-files` lists our models / textures / audio	Select ▾
No build (.apk / .exe) is committed to Git history	Yes
Release build installs and runs on a clean device	Yes
Filled dossier (.docx + .pdf) committed to docs/	Yes

3.1 Repository telemetry for your folder (run from the repo root; type the numbers)

Metric	Command	Value
Commits touching your folder	git log --oneline -- Group_13_Kartik_Soha_Atharv_Ojaswi/ | wc -l	1
Branches used by your group	git branch -r --list '*group-13*' | wc -l	1
Lines of C# code	find Group_13_Kartik_Soha_Atharv_Ojaswi -name '*.cs' | xargs cat | wc -l	11726
Total folder size (MB)	du -sm Group_13_Kartik_Soha_Atharv_Ojaswi	124
LFS-tracked files in your folder	git lfs ls-files | grep Group_13_Kartik_Soha_Atharv_Ojaswi | wc -l	18
Raw telemetry CSV files (real participants)	ls Group_13_Kartik_Soha_Atharv_Ojaswi/telemetry/*.csv | wc -l	17

3.2 Individual contribution (from git history; this table is checked against GitHub)

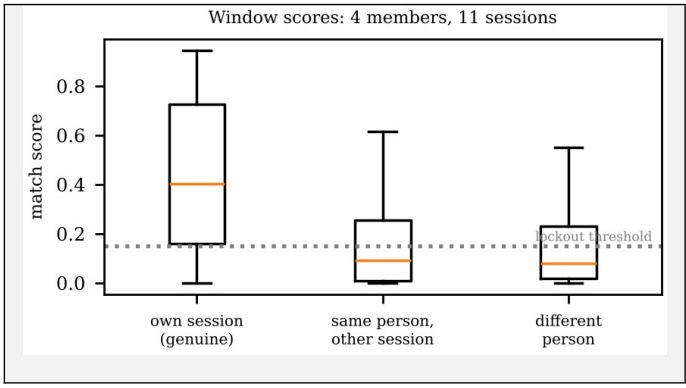
Member	GitHub username	Commits	% effort (agreed)	Main contribution (files / features)
Kartik Agrawal (I001)	kartik-agrawal-01	1	25%	type here
Soha Chand (I007)	type here	type here	type here	type here
Atharv Dixit (I013)	type here	type here	type here	type here
Ojaswi Gondalia (I019)	type here	type here	type here	type here

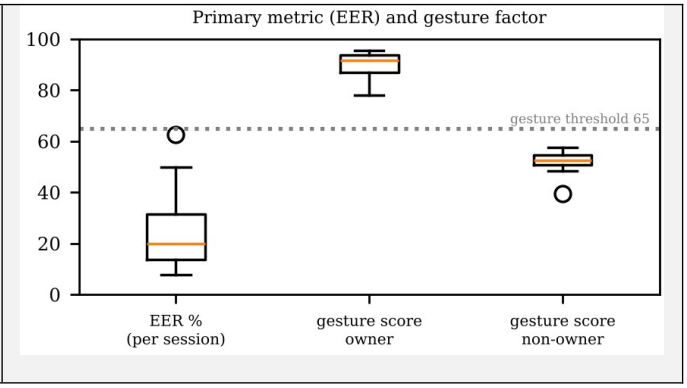
SECTION 4 · Human-in-the-Loop Usability Evaluation & Empirical Data Pack *DBL Usability Case Study*

Primary task metric	EER (%), FAR, FRR, ROC
Participants	Minimum N ≥ 10 real people (your PBL dossier recommends N = 10 genuine + 5 impostors). Voluntary, adult, written consent; anonymised IDs (P01, P02…).
Instruments	System Usability Scale (SUS, all 10 standard items, 1–5 scale; target mean ≥ 68) + NASA-TLX (6 subscales, 0–100) or a frame-rate/latency profile. Report Simulator Sickness (SSQ) if users wear a headset.
SUS scoring	Odd items: score − 1. Even items: 5 − score. Sum × 2.5 = SUS (0–100) per participant.

Real participants tested (N)	4 (team members), 11 sessions
Study design	One condition only
Testing dates (from – to) and place	26-27 Oct 2026, MPSTME Mumbai, Meta Quest 2
Raw CSV path in repo	Group_13_.../telemetry/real_participants/

Measure	N	Mean	SD	Median	Condition / comparison
SUS score	type here	type here	type here	type here	type here
NASA-TLX (raw)	type here	type here	type here	type here	type here
Primary task metric	11	25.4 %	17.4 %	20.0 %	Per-session EER (pooled 30.3 %)





DATA PROVENANCE DECLARATION (mandatory). Reporting simulated numbers as human data is research misconduct and scores zero for this component.

Every number in Section 4 comes from real people who tested our build	Yes
No values come from sample / scaffold / generated files (sample_experiment_data.csv, *_benchmark.csv, generate_paper_figures.py)	Yes
Every participant signed a consent form and we hold the forms	Select ▾
Our plots re-generate from the raw CSV in the repo	Yes

SECTION 5 · LinkedIn Video Showcase, AI Verification Audit & Peer Review *Studio Showcase & Viva Voce*

LinkedIn post URL	https://lnkd.in/p/d_G32T8Q
Video length (mm:ss)	3:54
Every member appears and speaks in the video	Yes

5.1 AI Interaction Audit Log: three cases where AI was wrong, made something up, or was unverifiable

Examples from this course: a fake 'Verizon 2024 XR' report, a fictional 'NEXORA Corp', a correct paper given the wrong journal. Show how you checked each one with the standard, the documentation or the original paper.

#	AI tool	What we asked & what AI said	How we caught it	What was actually true (with source)	Error type	Verdict
AI-01	Claude	Sim. of 40 users: EER 4.7 %, target met.	Ran / tested the code	Real Quest sessions: EER 17-30 %.	Wrong number or statistic	Corrected
AI-02	Other	Pfeuffer 2019 title: '…How They Look, Point and Walk'.	Checked the original source	'…Body Motion and Relations in VR', doi:10.1145/3290605.3300340	Real source, wrong details (author, year, venue)	Corrected
AI-03	Other	README: empirical N=50, EER 4.12 %.	Checked the original source	No such study (simulated). Real: N = 4.	Wrong number or statistic	Rejected

5.2 Peer review: watch both assigned LinkedIn videos and score them (0–2 per criterion)

Group reviewed	Problem	Build	Data	Honesty	Clarity	Best moment (mm:ss)	One question for the viva
Group 14: Ananya, Anvay, Mahit, Aryan	Select ▾	Select ▾	Select ▾	Select ▾	Select ▾	1:49	How did you connect your RL (SAC) model to Unity in real time?
Group 04: Rishi, Rahul, Srinidi	Select ▾	Select ▾	Select ▾	Select ▾	Select ▾	type here	type here

SECTION 6 · Individual Learning Record *not marked · one row per member · answer honestly*

These answers do not change your marks. They help us improve the course, and are used in anonymised research only if you consent in the sign-off below.

Member	Hours I spent (total)	My declared role matched my actual work (1 = not at all, 5 = fully)	Before this course I had used...	Before this course I had tested with real users	I learned a skill from a teammate of another programme	Share of my writing first drafted with AI	Wrong AI facts / citations I personally caught
Kartik Agrawal (I001)	About 60	5	Git only	Yes	Yes	0–25 %	0
Soha Chand (I007)	About 50	5	Git only	Yes	Yes	Select ▾	Select ▾
Atharv Dixit (I013)	50	5	Git only	Yes	Yes	Select ▾	Select ▾
Ojaswi Gondalia (I019)	type here	Select ▾	Select ▾	Select ▾	Select ▾	Select ▾	Select ▾

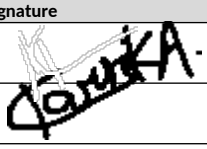
Member	The most valuable thing I learned (one line)	One thing I would change in this course (one line)
Kartik Agrawal (I001)	Real data humbled us: under 5% error in simulation, nowhere close on real people. Posture (sitting vs standing) mattered most.	type here
Soha Chand (I007)	Designing for continuous checking: most logins check you once at the door; we kept checking for the whole session.	type here
Atharv Dixit (I013)	Verifying AI output: it gave a 2019 paper with the wrong title, and we confirmed the real one by DOI and title.	type here
Ojaswi Gondalia (I019)	Testing on the headset caught an AI-code bug: room-coordinate features made joystick movement look like a different person.	type here

Track A: how this dossier becomes your 4-page IEEE paper

I. Introduction	Section 1: title, bottleneck, verified standard
II. Related work	Section 1.1: verified sources only (5+ with DOIs)
III. System design	Section 2: architecture, state machine, equations
IV. Evaluation	Section 4: protocol, N, SUS/TLX/metric, plots
V. Discussion & limitations	Sections 3–5: reproducibility (repo, release), AI audit lessons
Target venues (title sheet)	A: IEEE INDICON (IEEE India Council International Conference) · B: ACM AsiaCCS (ACM Asia Conference on Computer and Communications Security)

Academic Sign-off *every member answers and signs*

Answer each statement for yourself: (1) this work is original and every source and AI use is declared; (2) the code in our folder runs and the Release build installs; (3) I understand our whole project and can explain my own part in the viva voce.

Member (Roll)	(1) Original work	(2) Code runs	(3) Viva-ready	Signature	Optional research consent*
Kartik Agrawal (I001)	Yes	Yes	Yes		Select ▾
Soha Chand (I007)	Select ▾	Select ▾	Select ▾		Select ▾
Atharv Dixit (I013)	Select ▾	Select ▾	Select ▾		Select ▾
Ojaswi Gondalia (I019)	Select ▾	Select ▾	Select ▾		Select ▾

**Optional: may Dr. Nanade use anonymised data from this dossier and repository in education research publications? Your answer has no effect on your marks, and you can withdraw until results are published.*